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SECTION 01

PROJECT PHASE 3 VISUALIZATION & DASHBOARD

Analysing the Impact of Macroeconomic Indicators and Financial

Performance on Malaysian Banking Stocks

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1.0 Market Overview

The market overview dashboard consists of 6 visuals and 1 slicer as shown in Figure 1. This section provides detailed explanation and information of each visual, key insights as well as the business insights presented.

Sum of Performance Score by Stock (Bar Chart)

The performance score by stock provides a quick comparison of the overall performance of each banking stock. A bar chart is used because it allows easy comparison between categories and clearly highlights ranking differences. According to the chart, CIMB achieved the highest performance score of 5.1K, followed by RHBBANK at 4.2K and MAYBANK at 3.6K. This indicates that CIMB recorded the strongest overall performance among the three banks during the analysis period.

Trading Volume Trend by Year and Stock (Stacked Area Chart)

The trading volume by year and stock shows how actively each stock was traded over time. An area chart is used because it effectively displays trends and changes in trading activity across multiple years while emphasizing the magnitude of trading volume. The chart shows that RHBBANK experienced significantly higher trading volume between 2015 and 2018 compared to CIMB and MAYBANK. However, trading volume declined substantially after 2018. CIMB and MAYBANK maintained relatively stable trading volumes throughout the period, indicating more consistent market participation.

Average Price by Date (KPI Card)

The average price KPI provides a summary measure of the overall stock price performance. A KPI card is used because it allows users to quickly identify an important metric without analysing detailed charts. The dashboard shows an average stock price of 8.78, providing investors with a benchmark value for evaluating overall market performance within the selected period.

Open, High, Low and Close Price (Candlestick Chart)

The candlestick chart displays the daily opening, closing, highest and lowest prices of the selected stock. A candlestick chart is commonly used in financial analysis because it provides comprehensive information about short-term price movements and market sentiment in a single visual. The chart shows that prices fluctuated within a relatively narrow range, suggesting stable short-term market conditions with no extreme price swings during the selected period.

Closing Price by Date and Stock (Line Chart)

The closing price trend chart tracks stock price movements over time. A line chart is used because it is the most effective visualization for displaying continuous trends and comparing multiple stocks across a long period. The chart shows that MAYBANK consistently maintained the highest stock price among the three banks and reached approximately 12.4 in 2026. CIMB experienced a sharp decline around 2020 before recovering strongly and reaching approximately 9.0 in 2026. RHBBANK displayed steadier growth, increasing gradually to around 8.2 in 2026.

Risk-Return and Trading Activity Analysis by Stock (Bubble Chart)

The bubble chart analyses the relationship between stock volatility, average price change percentage and trading volume. A bubble chart is used because it allows three variables to be visualized simultaneously. The x-axis represents stock volatility, the y-axis represents average price change percentage and the bubble size represents trading volume. The chart shows that CIMB achieved the highest average price change percentage but also had relatively high volatility. MAYBANK exhibited lower volatility with moderate returns, while RHBBANK had the largest trading volume but a lower average price change percentage compared to CIMB.

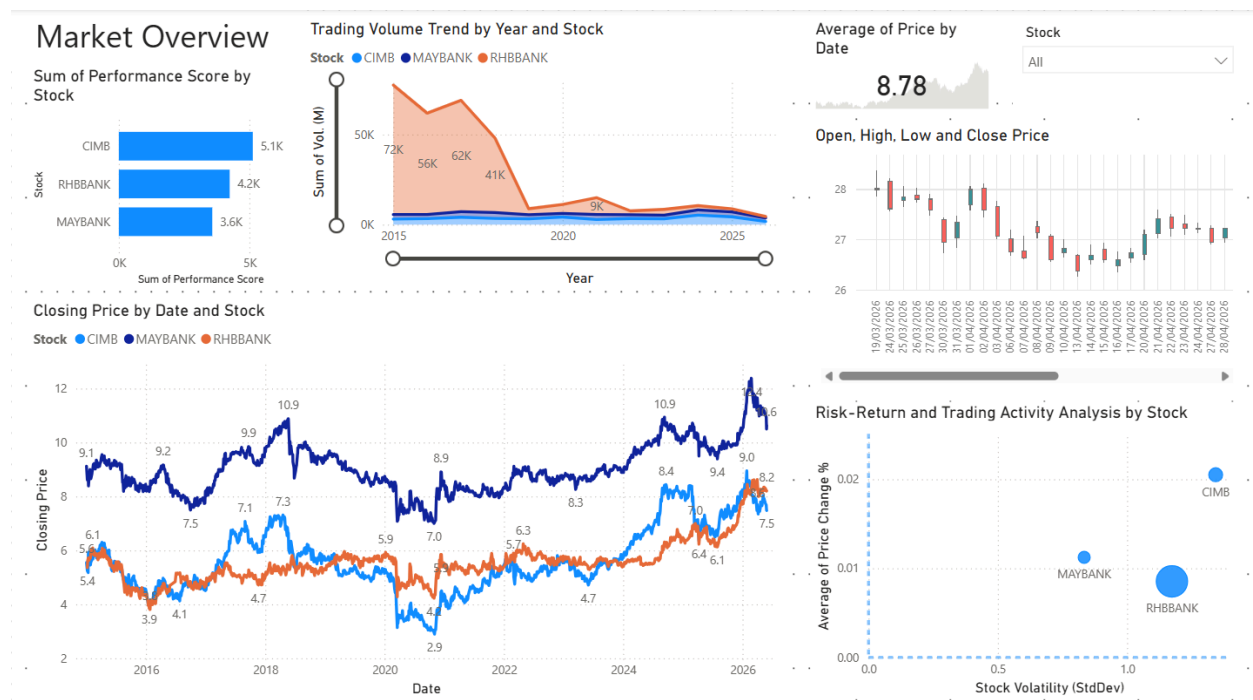


Figure 1 Market Overview Dashboard

Key Insights

Key Insight 1 – CIMB Demonstrates the Strongest Overall Performance

The dashboard reveals that each bank stock possesses a unique competitive advantage. CIMB recorded the highest performance score (5.1K) and achieved the highest average price change percentage, indicating the strongest overall performance and return potential among the three stocks. MAYBANK consistently maintained the highest closing price throughout the analysis period, increasing from approximately 9.1 in 2015 to 12.4 in 2026, demonstrating strong long-term capital appreciation. Meanwhile, RHBBANK recorded the highest trading volume, reflecting stronger market participation and liquidity compared to the other stocks. These findings suggest that CIMB is positioned as a high-return stock, MAYBANK as a long-term growth stock, and RHBBANK as a highly liquid stock suitable for active trading strategies.

Key Insight 2 – Risk and Return Differ Across the Three Stocks

The bubble chart highlights clear differences in risk-return profiles. CIMB generated the highest average price change percentage but also exhibited higher volatility. MAYBANK showed lower

volatility and more stable returns, while RHBBANK occupied a middle position with high trading volume but comparatively lower returns. This suggests that each stock appeals to different investor risk preferences.

Business Recommendations

Business Recommendation 1 – Align Stock Selection with Investment Objectives

Target Audience: Investors and Portfolio Managers

Investors and portfolio managers should select stocks based on their investment objectives and risk tolerance. CIMB may be suitable for investors seeking higher returns, as it recorded the highest performance score and average price change percentage. MAYBANK may be more appropriate for investors pursuing long-term capital growth due to its sustained price appreciation and relatively stable performance. RHBBANK may be preferred by investors who prioritize liquidity and flexibility, as its high trading volume allows for easier entry and exit from positions. A diversified portfolio that combines all three stocks could help balance growth opportunities, return potential, and market liquidity.

Business Recommendation 2 – Monitor Volatility Before Making Investment Decisions

Target Audience: Investors and Portfolio Managers

The dashboard demonstrates that higher returns are associated with higher volatility, particularly for CIMB Bank. Investors and banking institutions should regularly monitor volatility indicators alongside price performance when making investment decisions. Balancing return potential with risk exposure can improve long-term portfolio performance.

2.0 Fundamental Analysis

This dashboard analyses three Malaysian banking stocks, which are CIMB, MAYBANK and RHBBANK, across financial years 2016 to 2025 using the fact_stocks data model. Individual bank performance can be viewed by selecting a specific bank from the Stock slicer. It contains 4 KPI cards and 3 interactive slicers (Stock, Financial Year, Quarter) that filter all visuals at the same time. The EPS line chart displays all three stocks on the same chart, allowing easy side-by-side comparison. Revenue and Profit/Loss (P/L) are displayed using a common scale, making it easier to assess relative performance across the banks.

Figure 2 shows EPS of 19.77, Revenue of 8,917 million, P/L of 1,820 million and ROE of 2.72%. The first visual, Average EPS by Financial Year and Stock(Line Chart) shows the EPS trend from 2017 to 2025 for CIMB, MAYBANK and RHBBANK. All three banks experienced a noticeable decline during 2020-2021, reflecting the economic impact of the COVID-19 pandemic. Performance improved steadily afterward, with MAYBANK maintaining the highest EPS throughout the period.

Besides, Average Stock Price by Year (Line Chart) presents the stock price trend from 2016 to 2025. Throughout the period, MAYBANK maintained the highest share price, generally ragging between 8 and 10, while CIMB and RHBBANK traded within a lower range of 4 to 6. All three

banks experienced a noticeable decline in stock prices in 2020, reflecting the impact of the COVID-19 pandemic. However, prices gradually recovered in the following years and continued to improve towards 2025. When compared with the EPS trend, it can be observed that stock prices recovered more slowly than earnings during 2022–2023. This suggests that investors may have remained cautious and the market had not yet fully reflected the improvement in the banks' financial performance.

At the bottom left of the dashboard, Average Revenue (M) and Average P/L (M) by Financial Year (Clustered Bar Chart) compares Revenue and P/L for each financial year. Revenue shows a steady growth trend throughout the period, reaching its highest levels in 2024 and 2025. P/L follows a similar upward pattern, although at a smaller scale. The consistent relationship between Revenue and P/L suggests that the banks maintained relatively stable profitability while expanding their operations over time.

At the bottom right of the dashboard, the Fundamentals Table provides a detailed breakdown of each bank's financial performance by year. It displays key metrics including Average EPS, ROE, Revenue (M) and P/L (M) for every stock-year combination. The table is currently sorted in descending order by MAYBANK's results, highlighting its position as the highest-revenue bank across the observed period. This detailed view enables direct comparisons between banks and helps confirm the trends and patterns observed in the dashboard's graphical visualizations.

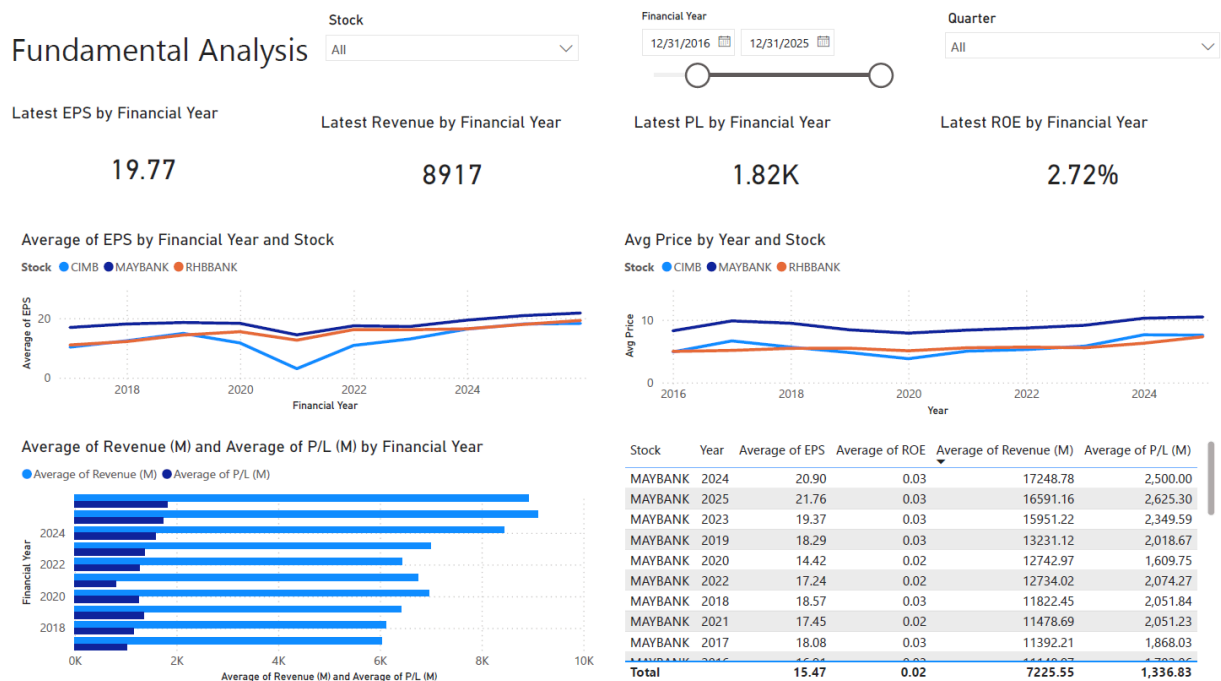


Figure 2 Fundamental Analysis Dashboard

Key Insights

Key Insight 1 - COVID-19 Significantly Impacted Bank Earnings in 2020–2021

The dashboard shows a significant decline in EPS across three banks during the COVID-19 period. CIMB experienced the largest drop with EPS falling from 14.90 in 2018 to just 3.04 in

2020. In comparison, RHBBANK demonstrated greater resilience, showing a less severe decline in earnings. This pattern highlights the sensitivity of the banking sector to major economic disruptions and confirms the widespread impact of the pandemic on financial performance.

Key Insight 2 - Earnings Recovered Strongly and Surpassed Pre-Pandemic Levels

Following the downturn, all three banks recorded a strong recovery in EPS. By 2023, CIMB achieved EPS of 16.63, MAYBANK reached 19.37 and RHBBANK recorded 16.45, all exceeding their respective pre-pandemic levels. The upward trend continues into 2024 and 2025, indicating sustained growth and improving profitability across the Malaysian banking sector. This recovery reflects stronger economic conditions and improved operational performance after the pandemic period.

Key Insight 3 - Stock Prices Recovered More Slowly Than Earnings

Although EPS returned to and exceeded pre-COVID levels by 2022, average stock prices remained below their 2017-2018 peaks during 2022 and 2023. This suggests that market sentiment and investor confidence recovered at a slower pace than company fundamentals. The gap between earning performance and share price movement may indicate that the market had not yet fully reflected the improved financial strength of these banks, potentially signaling undervaluation opportunities for investors during the recovery period.

Business Recommendations

Business Recommendation 1 - MAYBANK as the Strongest Long-Term Investment Choice

Based on the analysis, MAYBANK appears to be the strongest long-term investment option among the three banks. The company consistently recorded the highest EPS, ranging between 17 and 21, while revenue increased steadily from RM11,000 million to RM 15,951 million over the period. Combined with a stable ROE of 2.83%, these results reflect strong financial performance and reliable earnings growth, making MAYBANK an attractive choice for investors seeking stability and long-term value in the Malaysian banking sector.

Business Recommendation 2 - CIMB as a Potential Recovery Opportunity

CIMB presents an interesting opportunity for investors looking for growth potential. Although its EPS experienced the sharpest decline in 2020, falling to 3.04, the bank demonstrated a strong recovery by reaching 16.36 in 2023. This represents a remarkable 438% increase from its lowest point. If CIMB's share price has not fully reflected this earnings recovery, the stock could offer an attractive value investment for medium-term investors.

Business Recommendation 3 - Monitor the Relationship Between EPS and Share Price

The dashboard indicates that share prices across all three banks have not increased at the same pace as their earnings recovery. Investors and analysts should continue monitoring this gap throughout 2024 and 2025. If share prices begin to align more closely with improving EPS figures, it would support the view that the stocks are moving toward their fundamental value and could signal a stronger buying opportunity.

3.0 Macro Impact

The dashboard analyses how the macroeconomic indicator impacts the major Malaysian banking stock, namely MAYBANK, CIMB and RHBBANK. The dashboard enables users to explore historical trends and identify relationships between variables.

At the top of the dashboard, four cards present the latest values of the Overnight Policy Rate (OPR), Gross Domestic Product (GDP), Federal Funds Rate (DFF) and the Banking Stock Index.

A line chart displays the average stock prices of CIMB, MAYBANK and RHBBANK across the selected period. Another dual-axis line chart presents the trends of OPR, DFF and the Banking Stock Index over time, providing insights into the interaction between monetary policy and banking sector performance. A scatter plot illustrates the relationship between GDP and average stock prices, allowing users to assess whether economic growth is associated with changes in banking stock. Additionally, a scatter chart of OPR versus stock price change percentage is included to examine how stock returns vary under different interest rate environments.

The dashboard includes slicers on the top right corner. The Date slicer allows users to filter the analysis within a specific time range between 2015 and 2026, while the Stock slicer enables users to focus on individual banking stocks or compare multiple banks simultaneously.

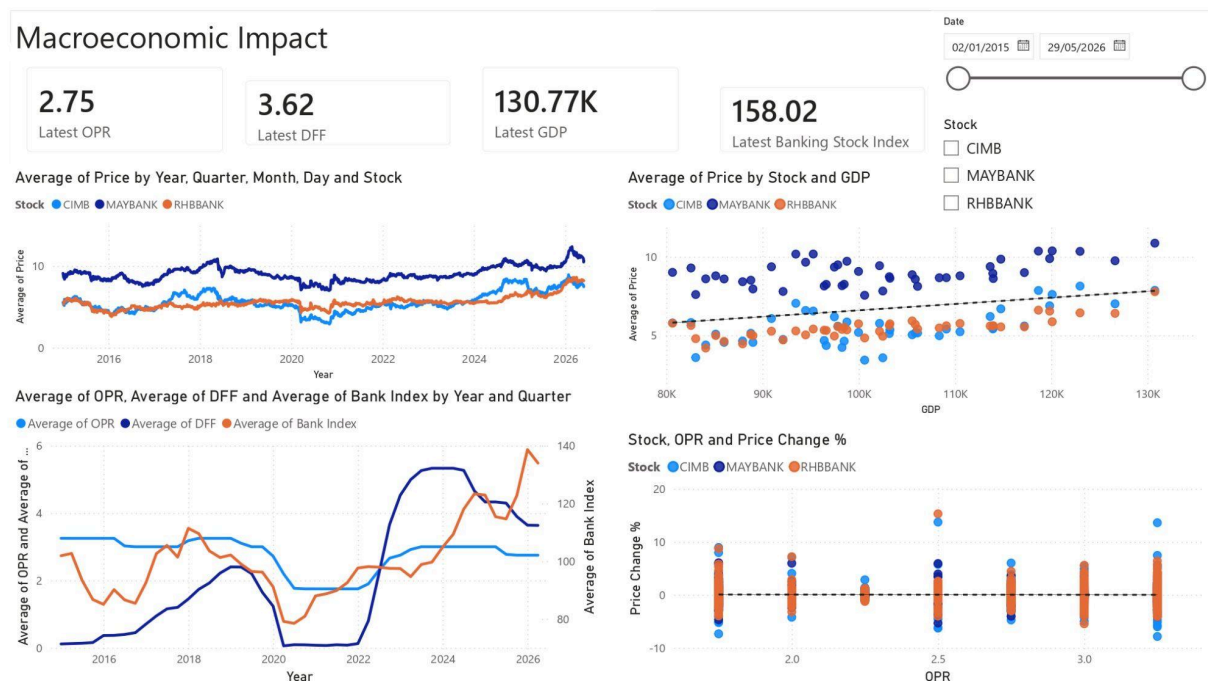


Figure 3 Macro Impact Dashboard

Key Insights

Key Insight 1 - Economic Growth Positively Influences Banking Stock Prices

The scatter plot between GDP and stock prices shows a positive relationship, as indicated by the upward trend line. Higher GDP levels generally coincide with higher bank stock prices,

suggesting that economic expansion improves banking sector performance through increased lending activity, investment and consumer spending. This relationship supports the view that GDP is a significant macroeconomic driver of bank valuations.

Key Insight 2 - OPR Changes Have Limited Immediate Impact on Stock Returns

The "Stock, OPR and Price Change %" analysis shows that stock price changes are dispersed across different OPR levels. Both positive and negative stock returns occur at the same OPR levels, indicating that interest rate changes alone do not directly determine short-term stock performance. Investors appear to react to broader economic conditions rather than solely to OPR adjustments.

Key Insight 3 - DFF and OPR Follow Similar Monetary Policy Trends

The Federal Funds Rate (DFF) and Malaysia's Overnight Policy Rate (OPR) generally move in the same direction over time. Significant rate increases occurred after 2022 as central banks responded to inflationary pressures. These synchronised policy movements reflect the influence of global monetary conditions on Malaysia's financial markets.

Key Insight 4 - Banking Sector Performance Improved Significantly After 2022

The Banking Stock Index rose strongly from approximately 2022 onwards, reaching its highest levels by 2026. This growth coincides with rising GDP levels, economic recovery after the pandemic, higher interest rate environments that may have supported bank profitability through improved net interest margins.

Business Recommendations

Business Recommendation 1 - Increase Exposure During Economic Expansion

Target Audience: Investors and Portfolio Managers

The investors and portfolio managers can monitor GDP growth indicators as an early signal for banking sector performance. Besides, they can consider increasing exposure to banking stocks during periods of sustained economic growth, as higher GDP is associated with stronger stock prices.

Business Recommendation 2 - Focus on Bank Fundamentals Rather Than OPR Alone

Target Audience: Investors

Investors should avoid making investment decisions based solely on OPR changes. Since stock returns remain mixed across different OPR levels, broader macroeconomic conditions would provide better predictive value.

Business Recommendation 3 - Strengthen Growth Strategies During Economic Recovery

Target Audience: Banking Institutions

Banking institutions shall expand lending activities in sectors benefiting from GDP growth. Besides, banking institutions can leverage periods of economic expansion to improve market share and profitability.